

AdaLN-Based Distance-Augmented Link Prediction in Graph Neural Networks

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Abstract—Link prediction is a fundamental task in graphs, where the goal is to predict missing links between nodes. Most existing approaches rely on node embeddings learned by graph neural networks and use simple decoders for link prediction, often overlooking global structural information such as the shortest-path distance between nodes. In this work, we propose a distance-augmented link prediction framework that incorporates shortest-path distance into link prediction. We encode distances using sinusoidal embeddings and use them to condition the decoder through adaptive layer normalization, allowing the model to adjust node representations based on structural proximity. Our experiments on the CORA dataset demonstrates that incorporating distance information leads to improved link prediction performance.

Index Terms—Link Prediction, Graph Neural Networks (GNN), GraphSAGE, Distance Encoding, Adaptive Layer Normalization (AdaLN).

I. INTRODUCTION

Link prediction is a fundamental problem in graph learning with applications in social networks, citation networks, recommendation systems, and biological networks. The task involves predicting if an edge exists between two nodes based on the observed structure of the graph and node features. With the success of graph neural networks (GNNs) [1], link prediction is approached using an encoder–decoder framework, where the encoder generates node embeddings and the decoder estimates the likelihood of a link between node pairs.

While significant effort has been devoted to designing more expressive encoders, the decoder in many existing approaches remains relatively simple, often relying on dot products or shallow multilayer perceptrons [1], [2]. As a result, these methods primarily depend on node embeddings and may fail to fully utilize important structural signals present in the graph.

One such signal is the shortest-path distance between nodes. Nodes that are closer in the graph are usually more likely to be connected than those that are far apart. However, this in-

formation is not incorporated in most standard link prediction models, as message passing in GNNs primarily captures local neighborhood information.

In this paper, we propose a distance-augmented link prediction framework that incorporates shortest-path distance into the decoding process. We first compute node embeddings using a GraphSAGE encoder [3]. We then encode the shortest-path distance between node pairs using sinusoidal frequency embeddings, which provide a smooth and expressive representation of distance. These embeddings are used to condition a decoder through AdaLN-Zero [4], enabling the model to modulate node representations based on structural distance before predicting links.

Our approach highlights the importance of combining node embeddings with global structural information for link prediction. We show that incorporating distance information into the decoder leads to more expressive models and improved predictive performance.

II. RELATED WORK

Graph Neural Networks (GNNs) have become the dominant approach for link prediction in graph-structured data [2], [5]. These models learn node representations by aggregating information from local neighborhoods [6]. One widely used method is GraphSAGE, which learns node embeddings by sampling and aggregating features from neighboring nodes [3].

Many link prediction methods rely on relatively simple decoders [1], [2]. As a result, the performance of these models often depends primarily on the quality of node embeddings, while the decoder plays a limited role in modeling structural relationships between node pairs.

A. Link Prediction Methods

Traditional link prediction methods include heuristic approaches such as Common Neighbors, Adamic–Adar, and Preferential Attachment, which rely on hand-crafted structural

features [7]. More recent learning-based methods use node embeddings generated by GNNs and predict links using similarity measures such as dot product, cosine similarity, or learned multilayer perceptrons [1], [2].

While these methods are effective, they typically rely only on node embeddings and do not explicitly incorporate global structural information such as shortest-path distance. As a result, they may fail to capture important structural patterns that influence link formation.

B. Adaptive Normalization and Feature Modulation

Adaptive Layer Normalization-Zero (AdaLN-Zero) was first introduced in the "Scalable Diffusion Models with Transformers" [4] paper as a means to incorporate information from heterogeneous embeddings (timestep and class embeddings). Precursors of AdaLN-Zero have also been used in deep learning to combine multimodal contextual information [8]. In these methods, embeddings containing external context are used to generate shift and scale parameters that modulate feature representations. This allows the model to adapt its behavior based on contextual information.

In this work, we apply a variant of AdaLN-Zero in the context of link prediction, where shortest-path distance is used as a conditioning signal to modulate node embeddings before decoding. This allows the decoder to adapt its predictions based on the structural relationship between nodes.

III. METHOD

A. Problem Setup

Let $G = (V, E)$ be a graph, where V is the set of nodes and E is the set of observed edges. Each node $v \in V$ is associated with a feature vector x_v . The goal of link prediction is to learn a function that predicts whether an edge exists between a pair of nodes (u, v) .

We follow the standard encoder-decoder framework for link prediction. An encoder $f(\cdot)$ maps each node to a low-dimensional embedding:

$$z_v = f(x_v, G)$$

where $z_v \in \mathbb{R}^d$ is the embedding of node v containing structural information.

A decoder $g(\cdot)$ then takes a pair of node embeddings and outputs a score representing the likelihood of an edge:

$$\hat{y}_{uv} = g(z_u, z_v)$$

The model is trained to assign high scores to observed edges and low scores to non-existent edges.

B. Node Embedding Encoder

To compute node embeddings, we use the GraphSAGE [3] encoder, which aggregates information from local neighborhoods. GraphSAGE updates node representations by aggregating features from neighboring nodes and applying a learnable transformation.

The node representations are updated as:

$$h_v^{(1)} = \sigma(W_1 \cdot \text{AGG}(\{x_u : u \in \mathcal{N}(v)\}))$$

$$h_v^{(2)} = W_2 \cdot \text{AGG}(\{h_u^{(1)} : u \in \mathcal{N}(v)\})$$

where $\mathcal{N}(v)$ denotes the set of neighbors of node v , $\text{AGG}(\cdot)$ is an aggregation function, W_1 and W_2 are learnable weight matrices, and σ is a non-linear activation function. Dropout is applied after the first layer to prevent overfitting.

The final node embedding is given by:

$$z_v = h_v^{(2)}$$

C. Shortest-Path Distance

In addition to node embeddings, we incorporate structural information in the form of shortest-path distance between node pairs. For a pair of nodes (u, v) , we compute the raw shortest-path distance $d_{raw}(u, v)$ in the graph. However, passing the raw distance poses problem causing the model to overfit.

Since very large distances are less informative for link prediction, we clamp the maximum distance to a fixed value D_{max} .

We also found that passing the distance embedding signal when $d = 1$ in its raw form caused overfitting because it implies there is an edge with probability one. Hence, we clamped the minimum distance to D_{min} . Hence, the distance used in our work is defined in eq. 1.

$$d(u, v) = \max(\min(d_{raw}(u, v), D_{max}), D_{min}) \quad (1)$$

This distance is then used as an additional input to the decoder.

D. Distance Encoding

Instead of using the raw shortest-path distance directly, we map the scalar distance value to a high-dimensional representation using sinusoidal embeddings. This allows the model to learn more expressive representations of structural distance.

Given a distance value d , we compute a sinusoidal embedding as:

$$\text{SE}(d) = [\cos(d \cdot \omega_1), \dots, \cos(d \cdot \omega_k), \sin(d \cdot \omega_1), \dots, \sin(d \cdot \omega_k)] \quad (2)$$

where the frequencies ω_i are defined as:

$$\omega_i = \exp\left(-\frac{\log(T) \cdot (i-1)}{k}\right)$$

The sinusoidal embedding is then passed through a multi-layer perceptron (MLP) to obtain the final distance embedding:

$$d_{emb} = \text{MLP}(\text{SE}(d)) \quad (3)$$

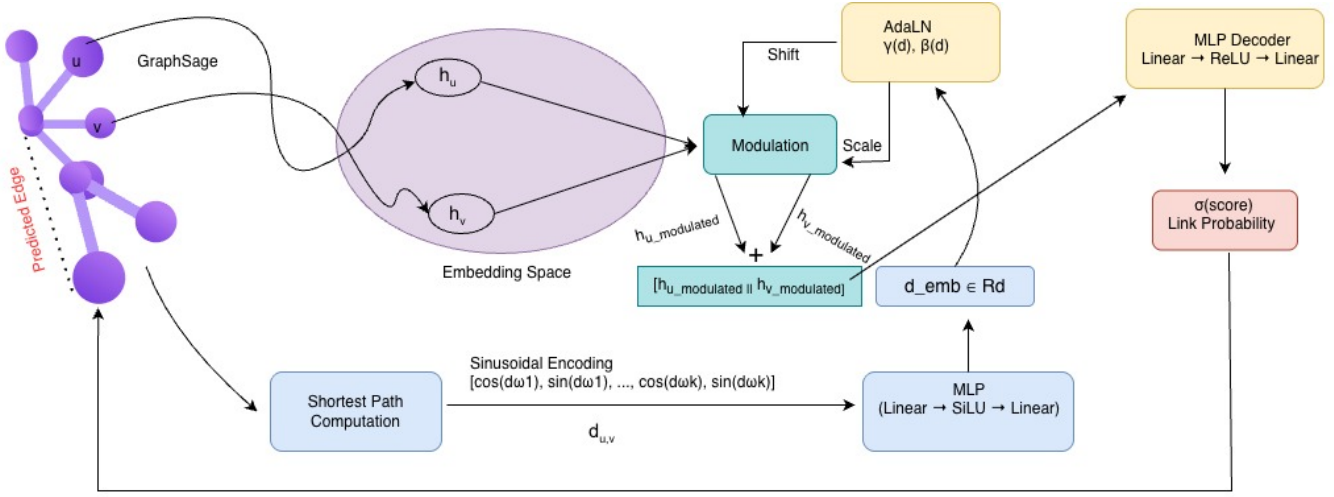


Fig. 1. Overview of the proposed method.

E. Distance-Aware Decoders

To evaluate the impact of node distance on link prediction, we consider multiple decoder architectures with increasing levels of distance integration.

1) *MLP Decoder*: As a baseline, we use a standard multi-layer perceptron (MLP) decoder that predicts links using only node embeddings.

Given embeddings z_u and z_v for nodes u and v , we concatenate them and pass them through an MLP:

$$h_{uv} = [z_u || z_v]$$

$$\hat{y}_{uv} = \text{MLP}(h_{uv})$$

This decoder does not incorporate any explicit structural information and relies entirely on the learned node embeddings.

2) *Raw Distance Concatenation Decoder*: To incorporate structural information, we extend the baseline decoder by including shortest-path distance as an additional feature.

Given the distance $d(u, v)$ between nodes, we concatenate it with the node embeddings:

$$h_{uv} = [z_u || z_v || d(u, v)]$$

$$\hat{y}_{uv} = \text{MLP}(h_{uv})$$

This approach allows the model to directly use distance information for link prediction. However, the distance is treated as a raw scalar, which may limit the strength of the distance signal.

3) *Distance Embedding Concatenated Decoder*: To increase the strength of the distance signal, we first convert the node distance into a high dimensional embedding as described by eq. 2 and eq. 3.

This embedding is then concatenated with the node embeddings to generate the input feature vector to the MLP

$$h_{uv} = [z'_u || z'_v || d_{emb}]$$

$$\hat{y}_{uv} = \text{MLP}(h_{uv})$$

This design has more signal strength of the shortest path distance but is too simple to capture the complex dependencies between distance, node embeddings and the likelihood of an edge between the nodes.

4) *AdaLN-Zero Distance-Conditioned Decoder*: We propose a distance-conditioned decoder that uses shortest-path distance to modulate node embeddings before predicting links.

First, the shortest-path distance $d(u, v)$ is transformed into a high-dimensional representation using the distance encoder as in eq. 3.

This embedding is then used to generate modulation parameters, namely shift and scale:

$$\text{shift, scale} = f(d_{emb})$$

The parameters of the final layer of the MLP used to calculate the modulation parameters are initialized to zero such that the AdaLN-Zero conditioning starts as a simple LayerNorm during training. These parameters are used to transform the node embeddings using AdaLN-Zero conditioning.

$$z'_u = \text{LN}(z_u) \cdot (1 + \text{scale}) + \text{shift}$$

$$z'_v = \text{LN}(z_v) \cdot (1 + \text{scale}) + \text{shift}$$

(where LN stands for Layer-Norm)

The modulated embeddings are then concatenated and passed through a decoder:

$$h_{uv} = [z'_u || z'_v]$$

$$\hat{y}_{uv} = \text{MLP}(h_{uv})$$

This design allows the model to adapt node representations based on structural distance before computing link likelihood. Unlike simple concatenation, this approach enables distance to influence the interaction between node embeddings in a more expressive manner.

F. Training Objective

The model is trained as a binary classification task to distinguish between existing edges and non-existing edges.

We have both positive (edge exists between the given nodes) and negative (no edge) samples in the training and test set. We use the binary cross-entropy loss with logits to train the model:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \sigma(s_i) + (1 - y_i) \log(1 - \sigma(s_i))]$$

where s_i is the predicted score for an edge, $y_i \in \{0, 1\}$ is the ground truth label indicating whether the edge exists, and $\sigma(\cdot)$ is the sigmoid function.

The encoder and decoder are trained jointly using the Adam optimizer.

IV. EXPERIMENTAL SETUP

A. Dataset

We evaluate our method on the Cora citation network dataset. In this dataset, nodes represent documents and edges represent citation links between documents. Each node is associated with a feature vector representing the document content.

The dataset contains 2708 nodes, 10556 edges, and 1433-dimensional node features. We perform link prediction by splitting edges into training and test sets.

B. Edge Split and Negative Sampling

We use a standard edge split where observed edges are divided into training and test sets. For training, we sample negative edges that do not exist in the graph using random negative sampling. The number of negative edges is chosen to be equal to the number of positive edges.

During evaluation, we compute predictions for both positive and negative test edges and evaluate performance using the Area Under the ROC Curve (AUC).

C. Implementation Details

The GraphSAGE encoder consists of two layers with hidden dimension 128 and output embedding dimension 64. Dropout with a rate of 0.5 is applied after the first layer.

The distance embedding module uses sinusoidal embeddings followed by a two-layer multilayer perceptron. The distance values are clipped to a maximum value of $D_{max} = 20$.

The model is trained using the Adam optimizer with a learning rate of 0.01 for 400 epochs.

V. RESULTS

We compare the performance of different decoders for the link prediction task.

TABLE I
LINK PREDICTION PERFORMANCE ON THE CORA DATASET

Model	AUC
MLP Decoder	0.77
Raw Distance Concatenation Decoder	0.83
Distance Embedding Concatenated Decoder	0.85
Distance-Conditioned AdaLN Decoder	0.88

The results show that incorporating shortest-path distance significantly improves link prediction performance. Furthermore, conditioning node embeddings using distance through adaptive normalization performs better than directly concatenating distance as a feature or as an embedding.

VI. ABLATION STUDY

To understand the contribution of different components of our model, we perform an ablation study by gradually adding distance information and analyzing its impact on link prediction performance.

A. Effect of Distance Information

We first evaluate how shortest-path distance affects link prediction. We compare three settings: (1) a baseline MLP decoder without distance information, (2) a decoder that concatenates raw shortest-path distance, and (3) a decoder that uses distance embeddings.

TABLE II
EFFECT OF DISTANCE INFORMATION ON LINK PREDICTION PERFORMANCE

Model Variant	AUC
MLP Decoder (No Distance)	0.77
Raw Distance Concatenation	0.83
Distance Embedding Concatenation	0.85

The results show that incorporating shortest-path distance significantly improves performance compared to using node embeddings alone. Furthermore, representing distance using embeddings performs slightly better than using raw distance values, indicating that high-dimensional distance representations are more expressive.

B. Effect of Distance Modulation

We next evaluate the impact of distance-conditioned modulation. Instead of directly concatenating distance embeddings with node embeddings, the AdaLN decoder uses distance embeddings to modulate node representations through learned shift and scale parameters.

The results show that distance-conditioned modulation performs better than simple concatenation.

C. Effect of Maximum Distance

Since very large shortest-path distances may not provide useful information for link prediction, we clip distances to a maximum value D_{max} . We evaluate the effect of different values of D_{max} .

TABLE III
EFFECT OF MAXIMUM DISTANCE THRESHOLD

D_{max}	AUC
10	0.85
20	0.88
50	0.83

We observe that clipping distances improves performance, as very large distances are less informative for predicting links.

VII. CONCLUSION

In this work, we studied the problem of link prediction in graphs and proposed a distance-conditioned decoding approach that incorporates shortest-path distance information into the link prediction model. While standard graph neural networks capture local neighborhood information through message passing, they do not explicitly model global structural relationships between nodes. To address this limitation, we introduced a decoder that conditions node representations on shortest-path distance using a technique inspired by AdaLN-Zero [4].

We explored multiple ways of incorporating distance information, including raw distance concatenation, distance embeddings, and distance-conditioned adaptive normalization. Our experiments show that incorporating structural distance significantly improves link prediction performance, and conditioning node embeddings using distance through AdaLN-Zero performs better than simple concatenation.

These results suggest that global structural signals such as shortest-path distance provide useful information for link prediction that is not fully captured by node embeddings alone. The proposed approach shows that conditioning mechanisms like AdaLN-Zero can be an effective way to integrate structural information into graph-based models.

In future work, this approach could be extended by exploring other structural signals such as random walk distances, common neighbor scores, or learned structural encodings. Additionally, more efficient methods for computing or approximating shortest-path distances could improve the scalability of the approach to larger graphs.

Overall, this work highlights the importance of combining node embeddings with global structural information for improving link prediction in graphs.

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